

JOSH HOMAN, MS, BSN, RN

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[Portfolio: clinical-analytics.vercel.app](#) | [GitHub: Joshdhoman](#) | [LinkedIn: linkedin.com/in/josh-homan](#)

PROFESSIONAL SUMMARY

Healthcare operations and data science professional with 12 years of clinical experience in critical care and interhospital patient placement plus a Master of Science in Data Science. Combines Epic workflow and patient-throughput expertise with Python, SQL, machine learning, statistical analysis, and data visualization. Builds reproducible, human-reviewed AI and analytics prototypes focused on patient flow, clinician experience, explainability, and operational efficiency.

TECHNICAL SKILLS

Data & Analytics: Python, SQL, R, pandas, NumPy, scikit-learn, statsmodels, statistical testing, logistic regression, predictive modeling

AI & Machine Learning: classification, decision trees, rule-based NLP / information extraction, model evaluation, feature engineering, cost-based threshold selection, explainability, human-in-the-loop workflows, rapid prototyping

Development & Clinical Systems: Streamlit, Tableau, Matplotlib, Jupyter, Git/GitHub, GitHub Actions (CI), pytest, PostgreSQL, AWS fundamentals, Epic EMR, patient throughput, transfer coordination, clinical workflow analysis

SELECTED ANALYTICS PROJECTS

Clinical AI Workflow Assistant for Patient Transfer Operations

Independent project | Synthetic data | Python, scikit-learn, rule-based NLP, Streamlit, pytest

- Built a decision-support prototype that converts unstructured interhospital referral notes into 16 evidence-linked clinical and operational fields, flags missing or conflicting information, and preserves source text for reviewer verification.
- Added an interpretable depth-six decision tree as a second opinion; held narrative-template families disjoint across train, validation, and test to reduce leakage. On 1,000 synthetic referrals, the held-out tree achieved 97.6% accuracy and 0.974 macro F1; deterministic rules delivered 1.000 macro precision on 73.2% coverage.
- Designed a human-in-the-loop Streamlit workbench with coordinator confirm/override, rationale capture, disagreement review, and clinical-routing safeguards; published a model card, data dictionary, CI tests, explicit limitations, and live demo. Synthetic data only; not validated for clinical use.

Patient Placement Analytics Dashboard

Independent project | Synthetic data | Python, pandas, Plotly, Streamlit

- Rapidly prototyped and deployed an interactive patient-placement dashboard using 6,000 reproducible synthetic adult encounters, translating frontline throughput questions into comparisons of transfer-center versus ED admissions by volume, care mix, level of care, and length of stay.
- Built filterable analytical workflows by admission date, hospital service, level of care, and patient class; added cohort-level KPIs, monthly trends, LOS distributions, and an encounter explorer with exact-cohort CSV export for traceable review.
- Validated dashboard calculations and filter behavior with automated tests, and documented metric definitions, assumptions, limitations, and a conceptual Epic-oriented data dictionary.

PROFESSIONAL EXPERIENCE

RN Clinical Placement Coordinator

Trinity Health | Maywood, IL | November 2018–Present

- Coordinate complex interhospital transfer and patient-placement workflows across a multi-hospital system, evaluating acuity, required services, physician acceptance, capacity, and available beds.
- Use Epic and real-time clinical information to support high-stakes placement decisions and identify recurring throughput and capacity bottlenecks under changing operational constraints.
- Translate clinical needs across referring physicians, receiving clinicians, nursing teams, and hospital operations to define placement requirements, resolve transfer barriers, and prioritize competing requests under changing operational constraints.

Registered Nurse, Bone Marrow Transplant Critical Care

Loyola University Medical Center | Maywood, IL | August 2014–October 2018

- Provided critical care for medically complex, immunocompromised patients with interdisciplinary teams and documented assessments and interventions in Epic.
- Served as BMTU liaison to the Nursing Informatics Governance Council, supporting clinical technology and workflow collaboration; mentored nursing students and new hires.

EDUCATION

Master of Science, Data Science

Eastern University | August 2026

Bachelor of Science, Nursing

Loyola University Chicago | May 2014

LICENSURE AND CERTIFICATIONS

Registered Nurse — Illinois

Google Data Analytics Professional Certificate — July 2024